

Errata and Supplement: adopting Unicode-aware chunking

v1.0.1 supplement. June 4, 2026.

Summary

This erratum does not replace the broad semantic map in the v1.0.0 release. In the old method, the token stream was split into fixed-length windows and each window was converted back to text on its own. Some chunk boundaries therefore cut through a character representation, producing the replacement character U+FFFD in the chunk text. This erratum evaluates a revised method that keeps the same 700-token windows and 100-token overlap while adjusting chunk boundaries to Unicode character boundaries.

U+FFFD appears because tiktoken tokens are not Unicode characters themselves; they are units over UTF-8 byte sequences. A CJK character is represented by multiple bytes and can, in some cases, span multiple tokens. When a fixed-length token slice is decoded by itself, the start or end of the slice may contain an incomplete UTF-8 sequence, which is then emitted as the replacement character U+FFFD.

In this corpus, the 265 revised chunks that reach 700 tokens have a median length of 458 characters, with a p10-p90 range of 432-489 characters. The 700-token window therefore corresponds roughly to about 460 CJK text characters in these classical Chinese-based materials. Character counts are Python string characters and include punctuation and symbols.

The 100-token overlap means that text near a boundary is also partly present in adjacent chunks. Boundary-level differences are therefore more likely to be averaged out in text centroids and volume-level summaries, which is one plausible reason for the broad stability observed here. The overlap does not, however, eliminate all chunk-level first-place source changes.

Technical note: the implementation uses tiktoken offsets to map token positions to Python string positions. The old behavior was reproduced with the same fixed-token decoding used in v1.0.0.

String Quality

method	chunks	chunks with replacement characters	replacement characters	median tokens	mean tokens
old method	433	256	326	700	692.935
revised method	433	0	0	700	692.679

Quantitative Evaluation

The embedding model is text-embedding-3-large. The old-method embeddings use the existing embedding cache. Only the revised chunks were newly embedded with the same model. Public outputs do not include raw text or chunk text.

comparison	count	min	p10	median	mean	p90	max
same-position chunks	433	0.89001	0.966485	0.997813	0.98916	1	1
text centroids	15	0.995906	0.996709	0.998511	0.99809	0.998773	0.999538
sect centroids	11	0.997908	0.997908	0.998751	0.998713	0.999069	0.999086

At the text level, the old and revised centroids remain closely aligned. The lowest text-centroid cosine is 0.995906.

text	ID	old/revised centroid cosine
佛說阿彌陀經	t0366_amida_sutra	0.995906
大樂金剛不空真實三麼耶經	t0243_rishu_kyo	0.996541
般若波羅蜜多心經	t0251_heart_sutra	0.996961

text	ID	old/revised centroid cosine
觀世音菩薩普門品	t0262_kannon_chapter	0.997215
妙法蓮華經	t0262_lotus_sutra	0.997908
稱讚淨土佛攝受經	t0367_praise_pure_land	0.998081
佛說觀無量壽佛經	t0365_meditation_sutra	0.998349
佛說無量壽經	t0360_larger_sukhavati	0.998511
金剛般若波羅蜜經	t0235_diamond_sutra	0.998672
大毘盧遮那成佛神變加持經	t0848_maha_vairocana	0.998693
大方廣佛華嚴經	t0279_flower_garland	0.998707
維摩詰所說經	t0475_vimalakirti	0.998728
解深密經	t0676_samdhinirmocana	0.998751
金剛頂一切如來真實攝大乘現證大教王經	t0865_vajrasekhara	0.998787
顯淨土真實教行證文類	kyogyoshinsho	0.999538

Three-Layer Check

The semantic layer here is an embedding-based source mixture: it uses cosine similarity between target chunks and source chunks. The source changes in this layer should therefore be read as changes in local source rankings derived from embeddings, not as a collapse of the embedding space as a whole.

The three-layer source-mixture map was also recomputed under the old and revised chunking methods. In the semantic source-mixture layer, the dominant source changes for 33 / 197 chunks. However, the dominant source by volume does not change. In the style/lexical layer, the dominant source changes for 1 / 197 chunks. In the source-marker layer, it changes for 0 / 197 chunks.

layer	chunks	median weight cosine	mean L1 distance	dominant-source changes	volume-dominant changes
semantic source-mixture layer	197	0.992013	0.123899	33 / 197	0 / 7
style/lexical layer	197	1	0.000657	1 / 197	0 / 7
source-marker layer	197	1	0	0 / 197	0 / 7

For the source-marker layer, the number of chunks with detected markers and the total number of marker hits are unchanged.

metric	old method	revised method
chunks with markers	99	99
total marker hits	170	170
exact marker-vector match rate	1	
chunks with changed marker vector	0	

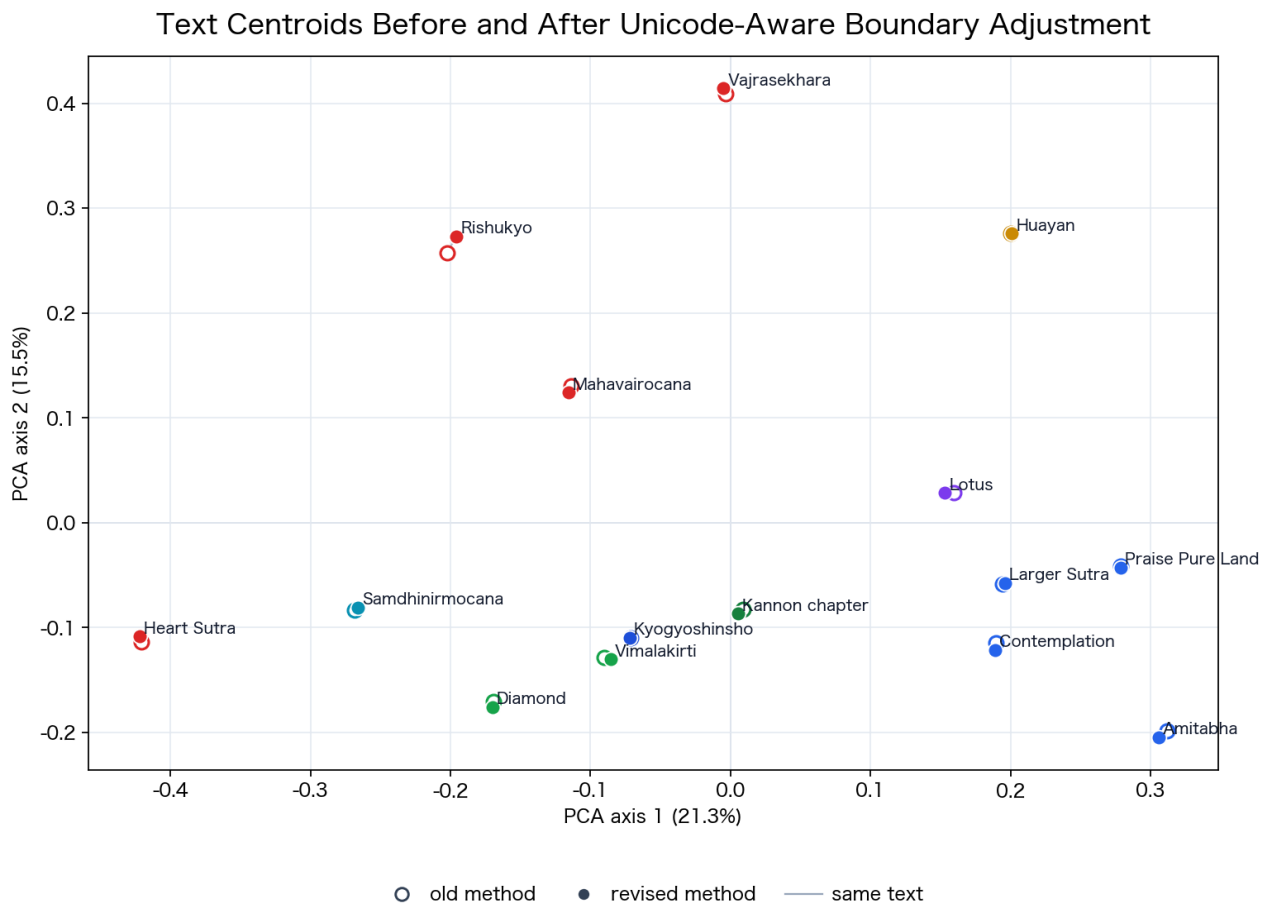


Figure: text centroids for the old and revised methods. Lines connect the same text.

Interpretation

The U+FFFF issue is a real text-quality and reproducibility problem. At the same time, embedding cosine for chunks at the same position remains high: mean 0.98916 and median 0.997813. Text and sect centroids also remain closely aligned. The broad semantic map in v1.0.0 is therefore not immediately invalidated.

The semantic source-mixture layer nevertheless shows measurable sensitivity in chunk-level first-place source assignments. This suggests that local source rankings can change under input noise when candidate sources are close. At the same time, the 100-token overlap plausibly spreads boundary-level differences into adjacent chunks and helps stabilize text centroids and volume-level tendencies. By contrast, volume-level dominant sources, the style/lexical layer, and the source-marker layer remain stable. This is also a methodological advantage of the three-layer design: it separates where the analysis is sensitive from where it is stable.

Displaying chunk excerpts, SAT line-range correspondence, individual-neighbor inspection, and future citation-level analysis should not use the old method as the baseline. Future work should use the revised Unicode-aware method.